CLASSICAL IMAGE PROCESSING FOR BIOMEDICAL IMAGES

Image Segmentation Without Machine Learning

Thresholding • Edges • Regions • Clustering • Statistical & Markov models • Deformable models

[Lecturer: Mykola Korabel

Learning Outcomes

- Derive 2D Convolution and Otsu's thresholding from first principles.
- Implement spatial filtering for noise reduction and edge detection.
- Utilize mathematical morphology for biological shape refinement.
- Design complete segmentation and feature extraction pipelines.

ROADMAP

Eight classical families, one storyline

1. Thresholding
2. Edges
3. Regions
4. Clustering
5. Statistics
6. Markov fields
7. Bias field
8. Snakes

No training data. No neural networks. Just intensity, geometry, and a few elegant ideas from the 1970s–1990s that still run in every medical-imaging pipeline today.

Why Classical CV?

Interpretability

Biomedical pipelines require accountability. Unlike deep learning "black boxes," classical operators have predictable, deterministic behaviors based on well-defined mathematical transforms.

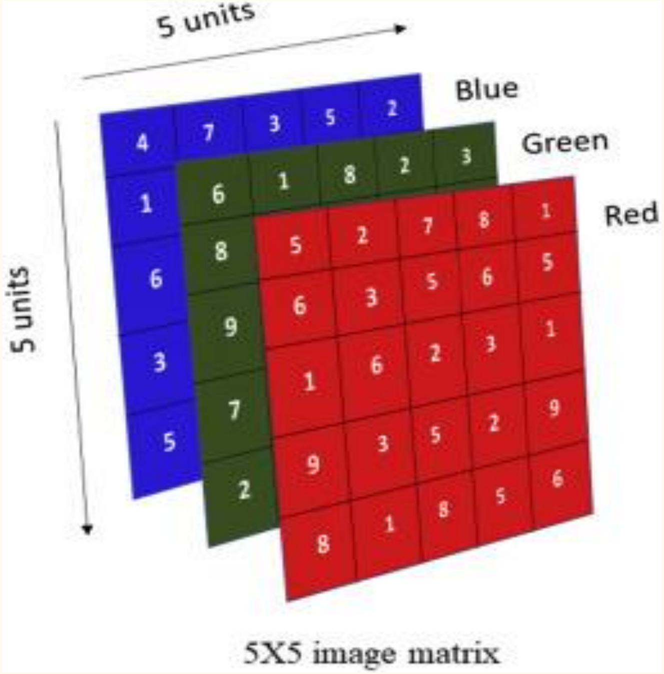
Auditability

For clinical diagnostic tools, the path from raw pixel to final measurement must be auditable. Classical methods provide clear intermediate steps (filtering, thresholding, morphology).

Images as Matrices

A digital image is a rectangular grid of discrete elements called **pixels**. Mathematically, it is a 2D matrix *I(x, y)* where each entry represents the light intensity.

$$I = \begin{matrix} p_{0,0} & \cdots & p_{0,W-1} \\ \vdots & \ddots & \vdots \\ p_{H-1,0} & \cdots & p_{H-1,W-1} \end{matrix}$$



RGB Channels

Red

Long wavelengths.

Green

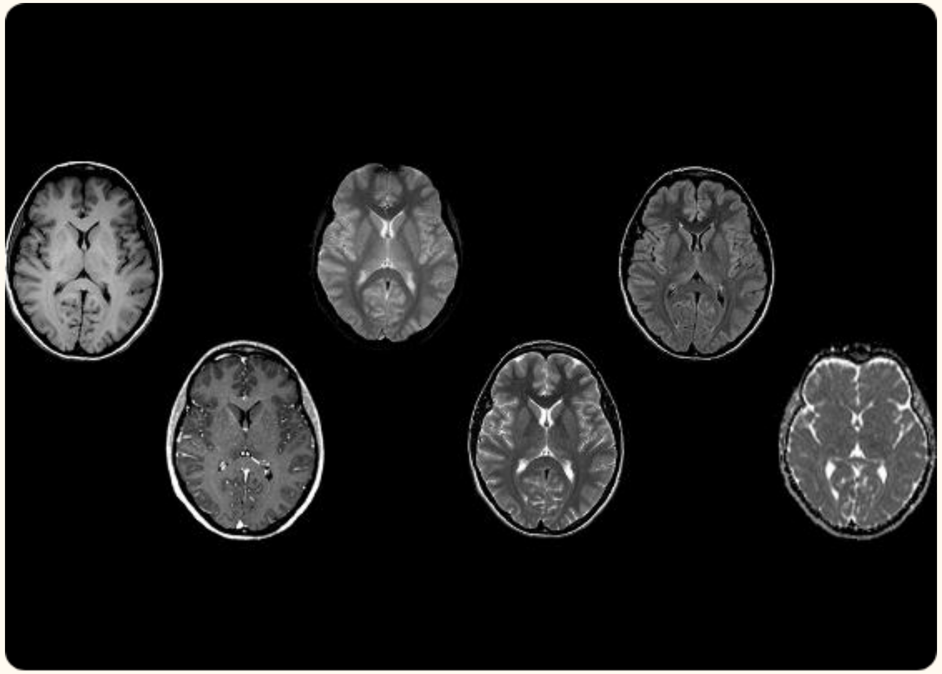
Mid wavelengths.

Blue

Short wavelengths.

Colour images are stacks of three 2D matrices. In histology (H&E), channels separate the eosin (red/pink) and hematoxylin (purple/blue) stains.

Grayscale Intensities



Standard grayscale images use 8-bit integers (unsigned integers uint8).

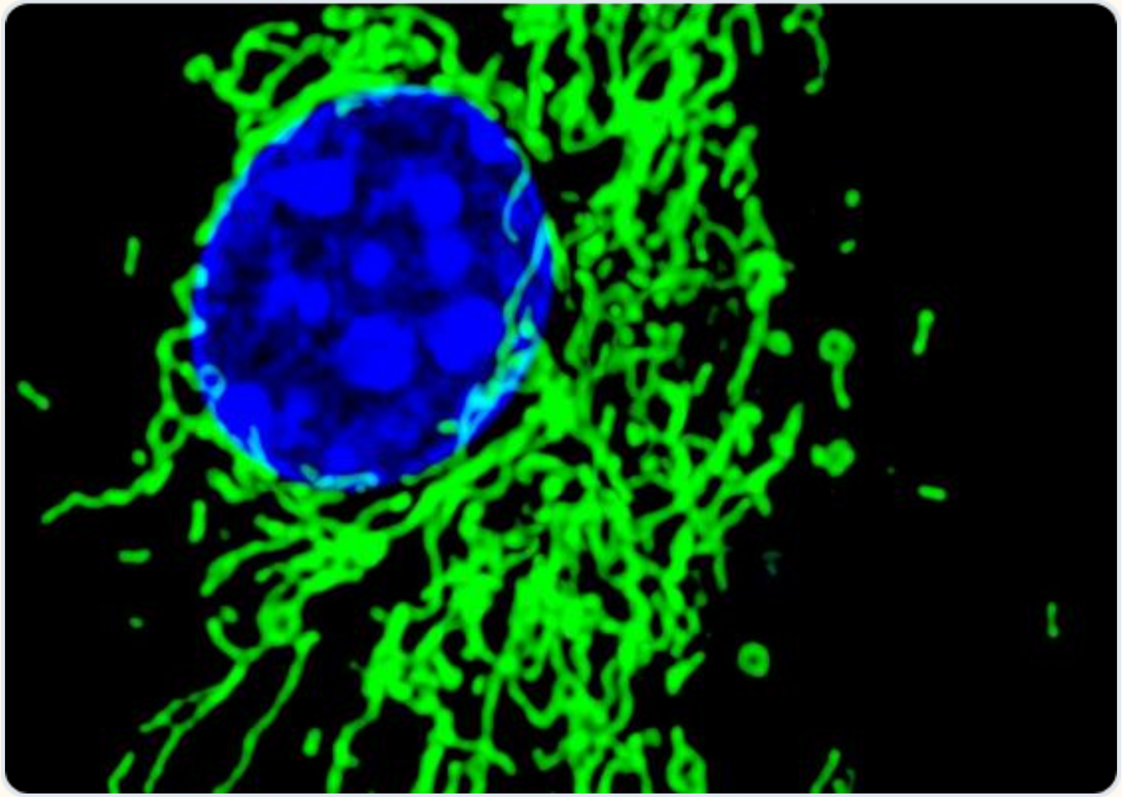
Values range from **0 (Black)** to **255 (White)**.

$255 = 2^8 - 1$

Most medical imaging uses higher bit-depths (uint16, uint32, uint64) for subtle tissue differentiation.

$2^{64} - 1$ (18,446,744,073,709,551,615)

The Alpha Channel



Transparency & Masks

The 4th channel (RGBA) encodes transparency. In biomedical workflows, this is often used to overlay segmentation masks (labels) over raw anatomical data without obscuring it.

Image Formats: Formats like JPEG do not support transparency. You should use Portable Network Graphics (PNG) to maintain an alpha channel.

Medical Bit Depth

Bit Depth	Unique Values	Use Case
8-bit	256	Web displays, histology PNGs.
12-bit	4,096	Many Digital X-ray systems.
16-bit	65,536	High-end CT and MRI (DICOM).

Higher bit depth reduces "quantization error" and allows radiologists to distinguish subtle variations in soft tissue density.

NumPy Representation

Array Shape

An RGB image is a NumPy array of shape (H, W, 3). Grayscale is (H, W). Note that indices are (row, column), corresponding to (y, x) coordinates.

Data Types

Ensure your arrays are uint8 for display or float64 for calculations. Normalized floats (0.0 to 1.0) prevent overflow errors during mathematical operations.

Exercise 1: Image Dimensions

What is the total number of pixels in a 512×512 RGB image?

Think about the Height, Width, and Channel depth.

Solution 1

Total Pixels in one colour channel: multiply the width and height of the image

512 * 512 = 262,144 pixels

$$512 \times 512 \times 3 = 786{,}432 \text{ values}$$

The NumPy shape is (512, 512, 3). This reflects the row-major storage of the three colour planes (Red, Green, and Blue).

Biomedical Formats

PNG

Lossless compression. Good for processed results.

TIFF

High bit-depth support (16-bit). Standard for microscopy.

DICOM

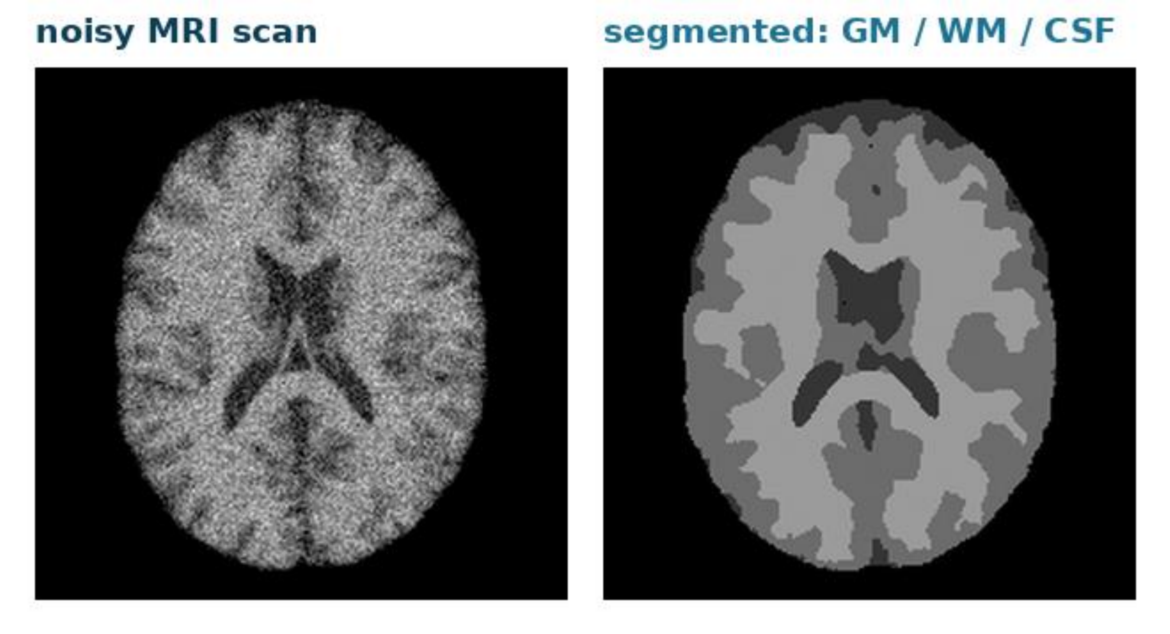
Clinical standard. Stores metadata + images.

Why DICOM Matters

DICOM (Digital Imaging and Communications in Medicine) combines image data with critical metadata: **pixel spacing** (mm), patient orientation, and clinical parameters.
Without pixel spacing, we cannot calculate the *physical* size (e.g., area in mm^2) of a tumor or organ.

What is segmentation, really?

- Segmentation = dividing an image into regions that are "homogeneous" in some sense (intensity, texture, colour).
- Classification = assigning every pixel a label from a known, agreed set (e.g. grey matter, white matter, CSF, air).
- The two are two sides of one coin: classify every pixel → you get a segmentation for free, and vice versa.



Same slice — raw intensities vs. a tissue-class labelling

KEY IDEA Every method in this lecture is really just a different way of answering "which pixels belong together?"

Why biomedical segmentation is hard

Noise

~10% noise-to-signal in MRI — about 10× worse than a phone photo.

Partial volume effect

Many voxels straddle two tissue types: poor sampling blends their intensities.

Complex texture & shape

Real organs and lesions rarely look like clean geometric blobs.

Subtle signs

The clinically interesting differences are often the smallest ones.

KEY IDEA "Brain MRI is as easy as it gets" — if it's hard here, expect it to be harder on pathology, ultrasound, or histology.

A taxonomy of classical methods

Pixel-based

Threshold each pixel by its intensity alone

Otsu, adaptive T

Edge-based

Find boundaries as sharp intensity changes

Sobel, Canny

Region-based

Grow / split / flood regions of similar pixels

Region growing, watershed

Clustering

Group pixel features without spatial structure

k-means

Model-based

Add a statistical / spatial prior

Bayes, MRF, bias field

Deformable models

Evolve a curve toward object boundaries

Snakes, level sets

PART 1 OF 8

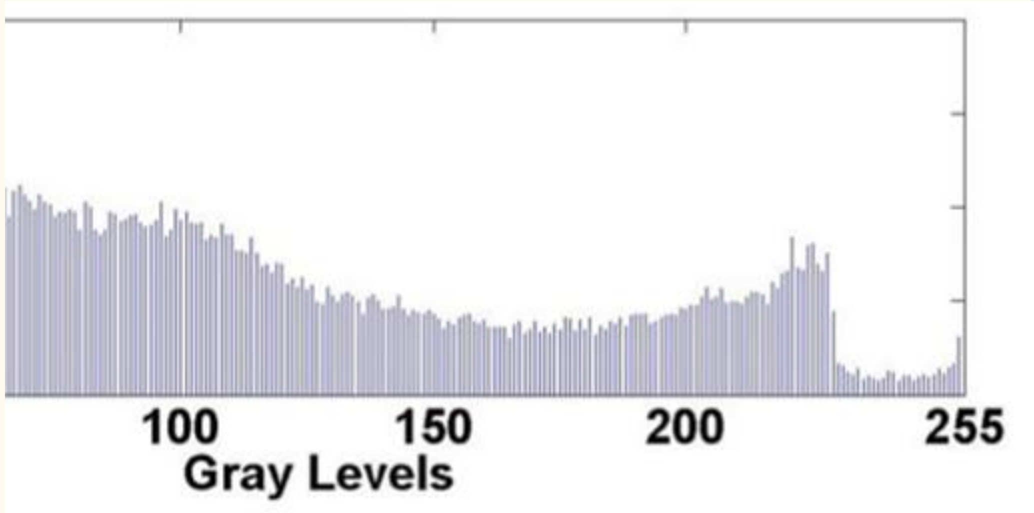
Thresholding

The simplest possible segmentation: one number splits the whole image.

Histograms

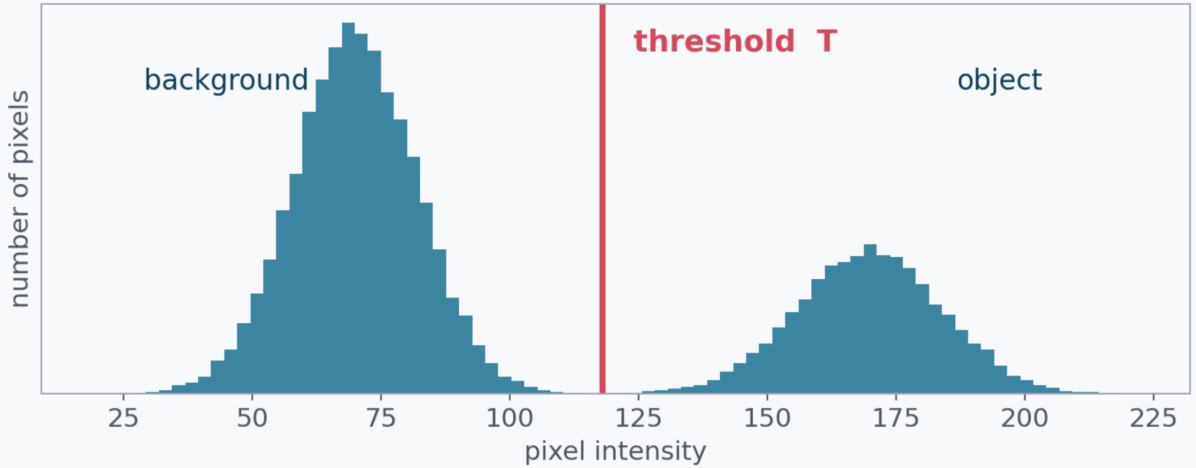
A histogram counts the frequency of each intensity level. It reveals if an image is underexposed, overexposed, or lacks contrast.

Peaks in the histogram often represent distinct tissue types (e.g., bone vs. soft tissue).



Global thresholding

- Pick one threshold T. Every pixel above T → object; below → background.
- **$g(x,y) = 1$ if $f(x,y) > T$, else 0**
- Works when the histogram is bimodal — one bump per class.
- **The only real question: how do we choose T objectively?**

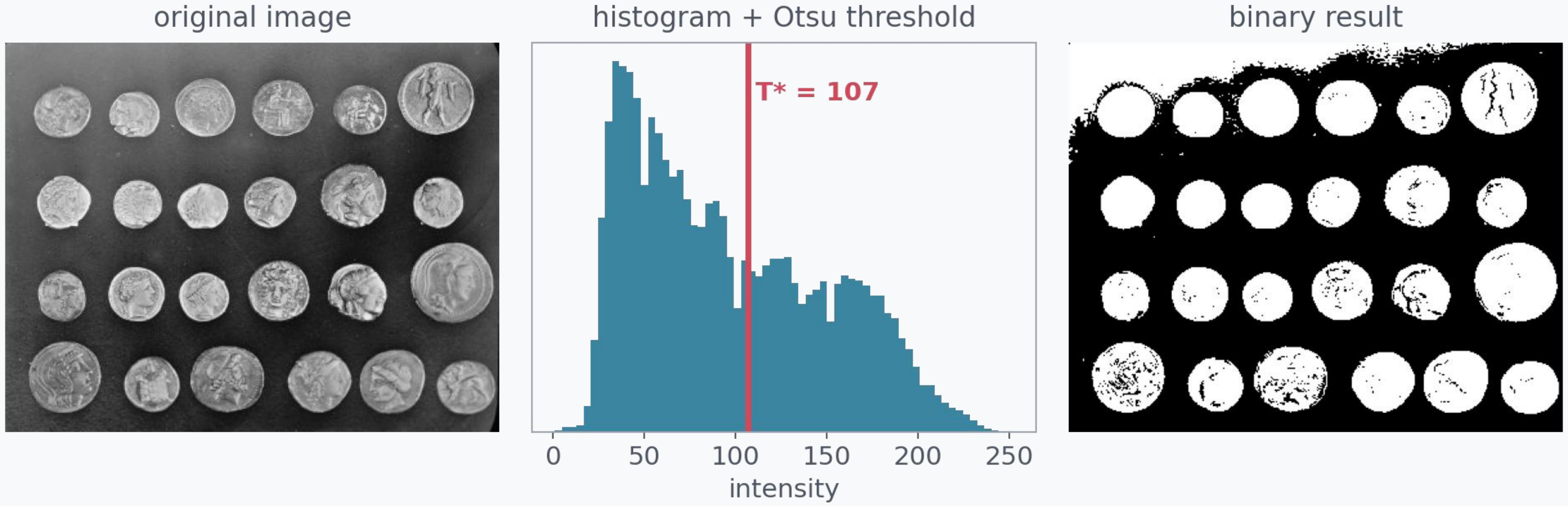


Two overlapping populations — background vs. object intensities

KEY IDEA Thresholding only needs the histogram, not the image layout — that's its strength and its blind spot.

Otsu's method: choosing T automatically

- Try every possible T. For each one, split pixels into two classes.
- Pick the T that makes the two classes as different from each other as possible (and as tight internally as possible).
- **No derivative, no calculus — just a search over ~256 candidate values.**



Otsu finds T automatically from the histogram shape — no manual tuning

Otsu’s Method

Optimal

Automatic Threshold Selection

Otsu’s algorithm automatically calculates a threshold by **maximizing between-class variance**. It treats the histogram as two distributions and finds the best point to separate them.

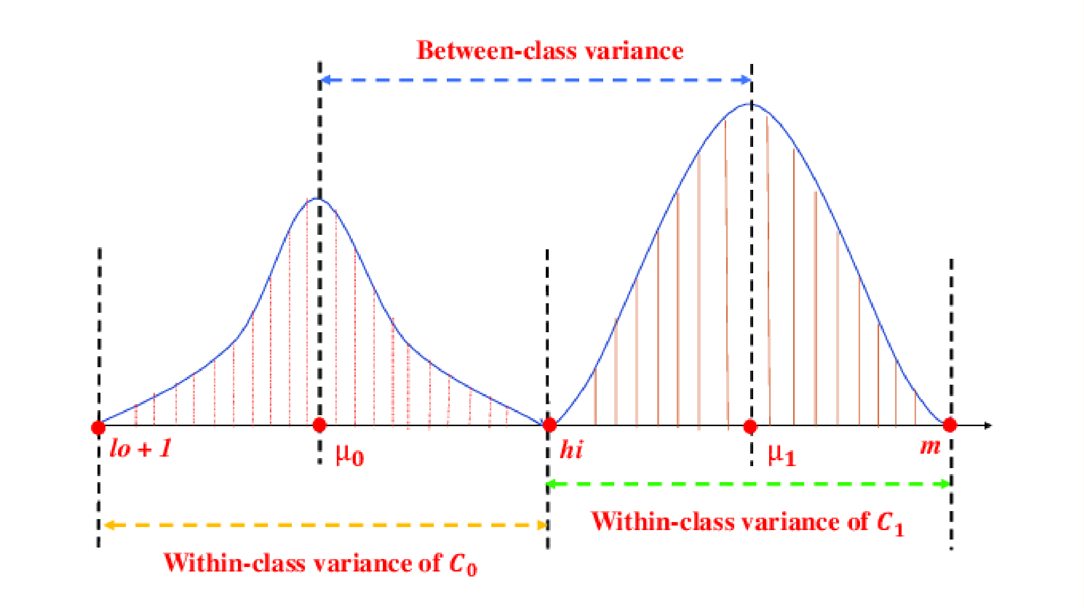
Otsu Derivation: Probability

Let $p(i)$ be the probability of intensity i. For a threshold k, the cumulative class probabilities for background and foreground are:

$$\omega_0(k) = \sum_{i=0}^{k-1} p(i), \quad \omega_1(k) = \sum_{i=k}^{L-1} p(i)$$

ω_0 is the background weight and ω_1 is the foreground weight.

Otsu Derivation: Variance

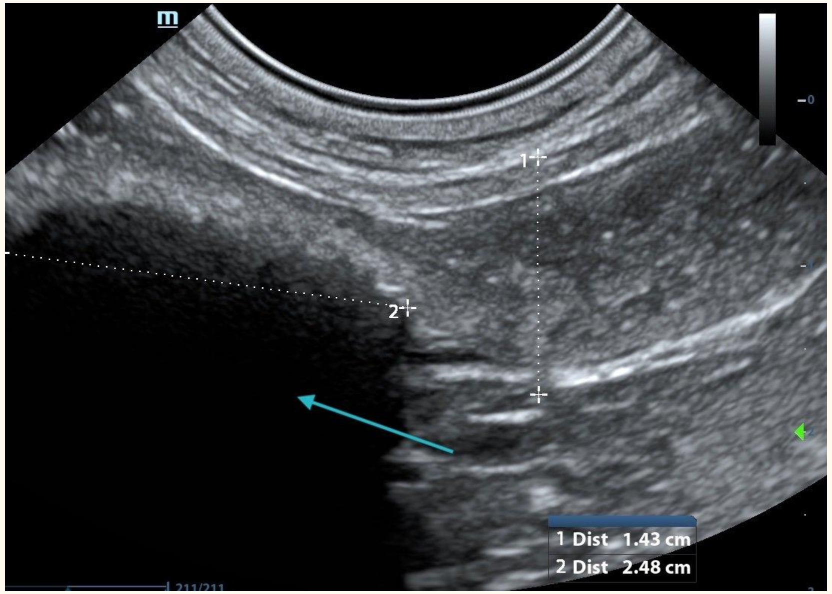


The goal is to maximize the between-class variance:

$$\sigma_b^2(k) = \omega_0(k)\,\omega_1(k)\,[\mu_0(k) - \mu_1(k)]^2$$

By iterating through all possible values of *k* (0 to 255), we find the one that yields the highest separation between the two means.

Exercise 4: Strategy



If your medical scan has severe "shadowing" on the left side, would Otsu's method work well?

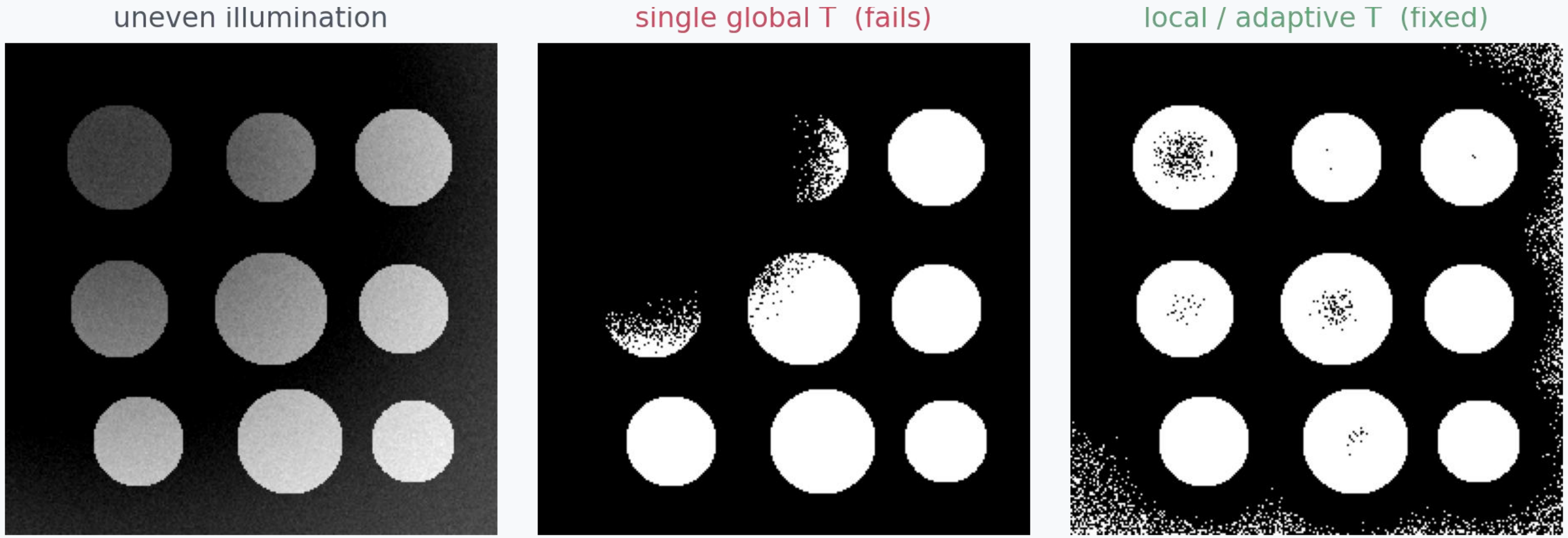
Solution 4

Probably Not

Otsu is a **global** method. Shadowing would shift the intensities of objects on the left into the "background" range of the right, leading to missing objects. **Adaptive thresholding** would be required.

The illumination problem

- Real scanners and microscopes don't illuminate evenly — a single global T can't fit both ends of a brightness gradient.
- Fix: compute a local threshold inside small windows, sliding across the image — “adaptive” or “local” thresholding.



KEY IDEA Uneven illumination is the small-scale cousin of the MRI “bias field” we'll meet again in Part 7.

Recap: thresholding

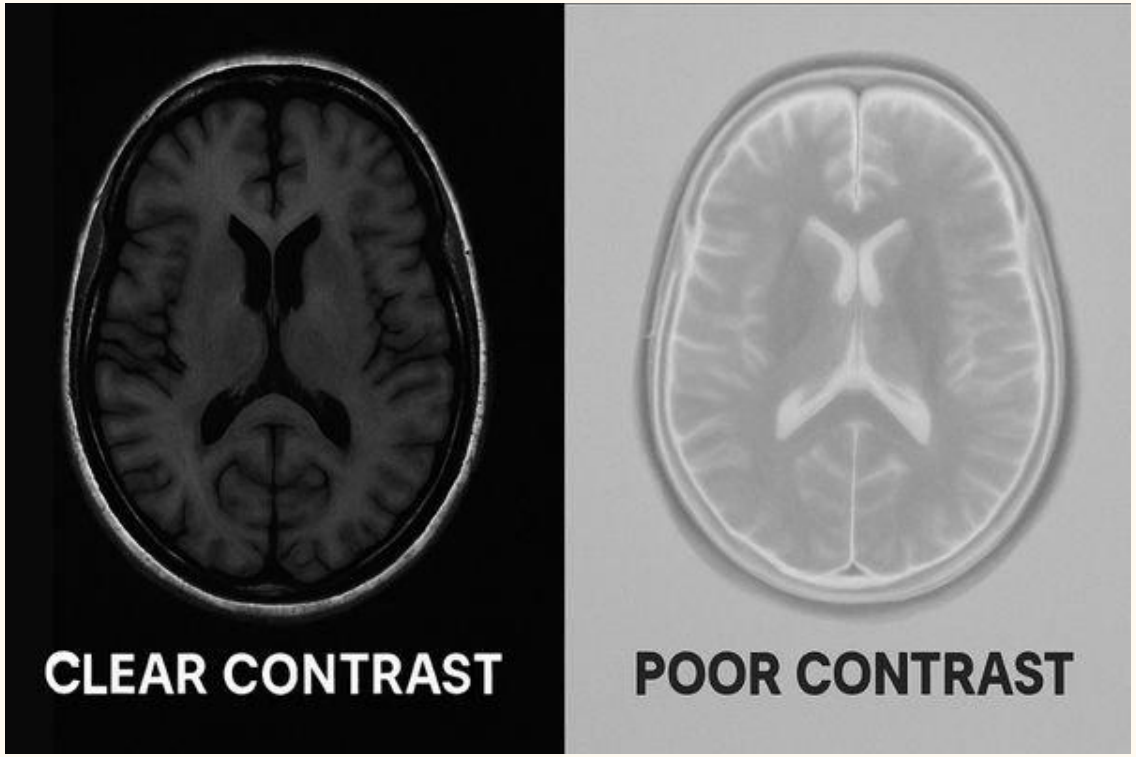
- Fast, simple, only needs a histogram — always worth trying first.
- Otsu removes the guesswork from choosing T.
- Ignores spatial position entirely — a noisy pixel far from any real object is judged purely on its own intensity.
- Fails under uneven illumination unless you go local / adaptive.

KEY IDEA Thresholding answers “what intensity?” — it never asks “where?”. Everything from here on adds some notion of “where”.

Contrast Issues

Medical images often suffer from **low contrast** due to limited exposure (to save radiation) or similar tissue densities.

In the adjacent image, the narrow intensity range makes it difficult to see subtle features in the dark background.



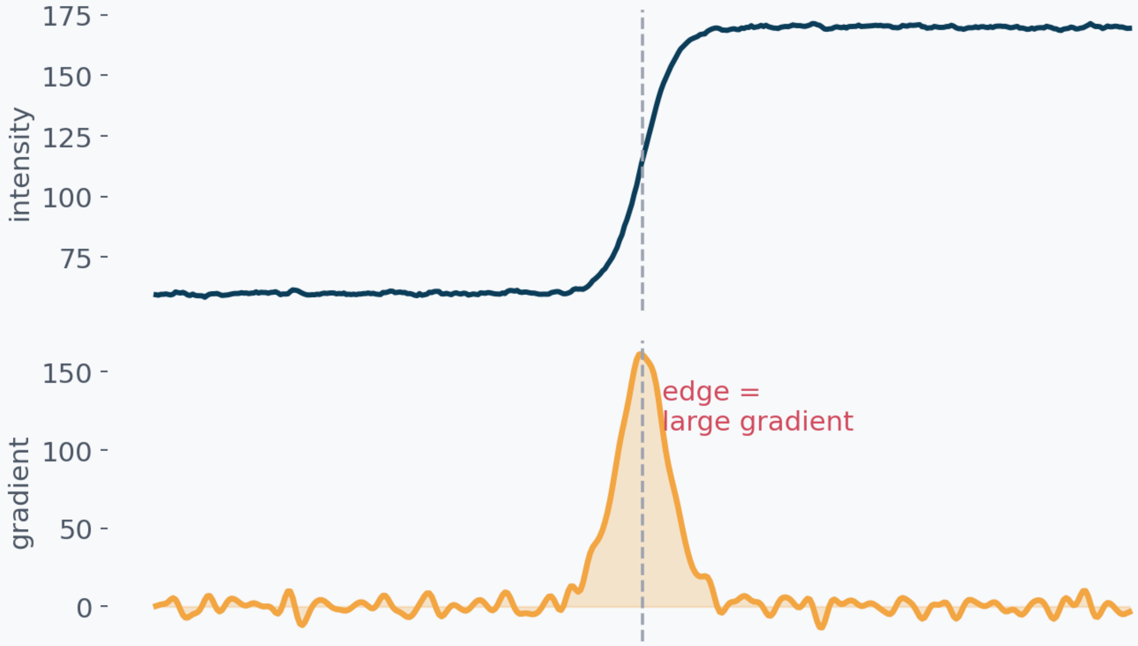
PART 2 OF 8

Edge-based segmentation

Stop asking “what class is this pixel?” — ask “is there a boundary here?”

The idea: edges are sharp intensity changes

- Walk along a line through the image and plot intensity — a boundary shows up as a steep ramp.
- The derivative (gradient) of that ramp has a sharp peak exactly at the edge.
- **So: edge detection = “where is the gradient large?”**



A 1-D intensity profile crossing a boundary, and its gradient

Gradient operators: the Sobel kernel

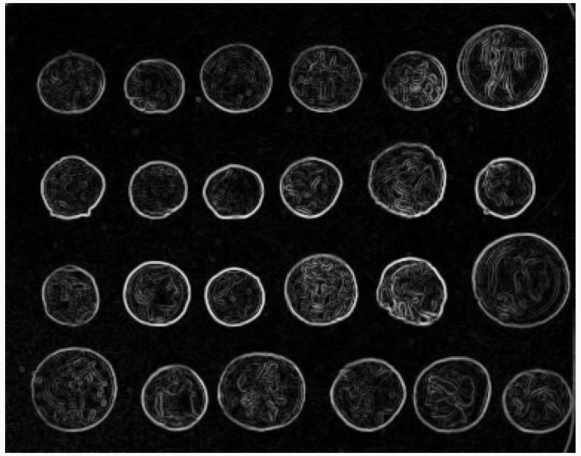
- In 2-D, slide a tiny 3×3 weighted window over the image to estimate the local gradient in x and y.
- **Sobel-x kernel:**

-1	0	1
-2	0	2
-1	0	1

Combine the magnitude of the x- and y- gradients → strong response exactly where intensity changes fastest, in any direction.

original

Sobel gradient magnitude



Sobel highlights every boundary — including coin texture and noise

Canny: a cleaner edge map

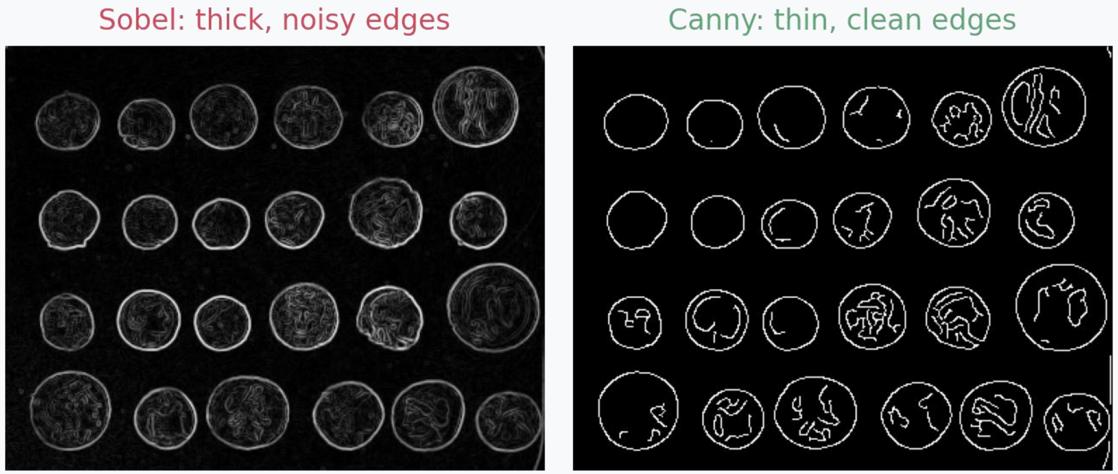
1
Smooth (reduce noise)

2
Compute gradient

3
Thin to 1-pixel-wide ridges

4
Keep only strong, connected edges

Four simple steps — no new mathematics, just smoothing, gradients, thinning, and a smart double threshold ("hysteresis") that keeps weak edges only if they connect to a strong one.



Sobel's thick, noisy response vs. Canny's thin, clean boundaries

The catch: edges aren't regions

- Edge detectors mark boundary pixels — they don't tell you what's inside vs. outside.
- Real edge maps have gaps (weak contrast), spurs, and branches.
- Turning a broken outline into a closed region usually needs extra work: linking, morphological closing, or a completely different strategy.

KEY IDEA If you need a filled region, not just a boundary, region-based methods often get there more directly.

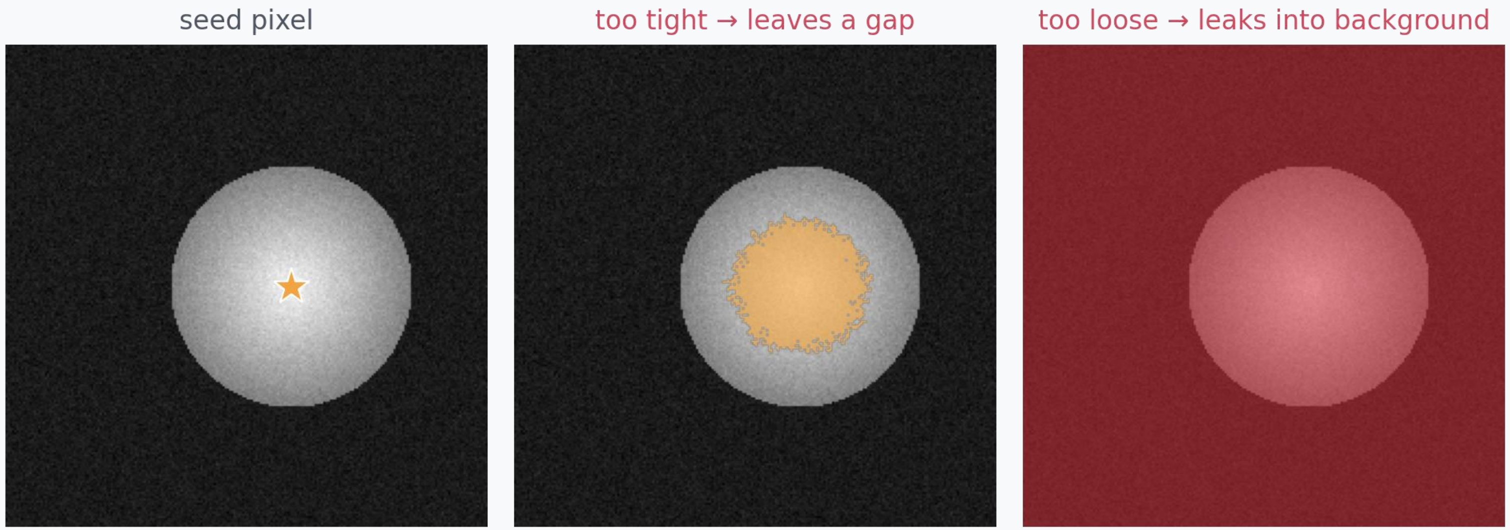
PART 3 OF 8

Region-based segmentation

Build regions directly, pixel by pixel, instead of chasing boundaries.

Region growing

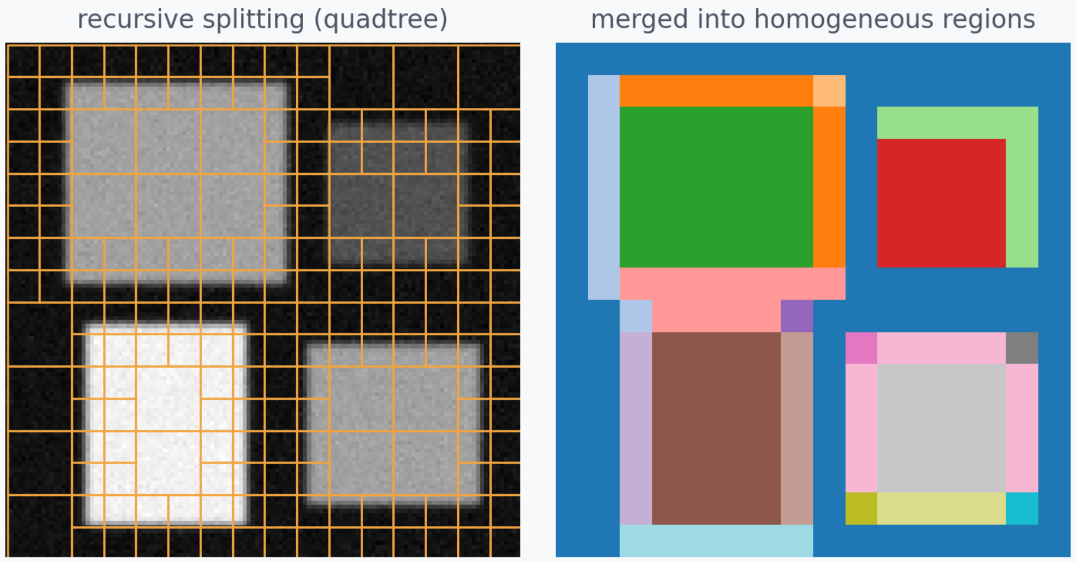
- Start from one seed pixel inside the object of interest.
- Repeatedly add any neighbouring pixel whose intensity is close enough to the seed (within a tolerance).
- Stop when no more neighbours qualify.



KEY IDEA One parameter — the similarity tolerance — controls the entire trade-off between under- and over-growing.

Split & merge (quadtree)

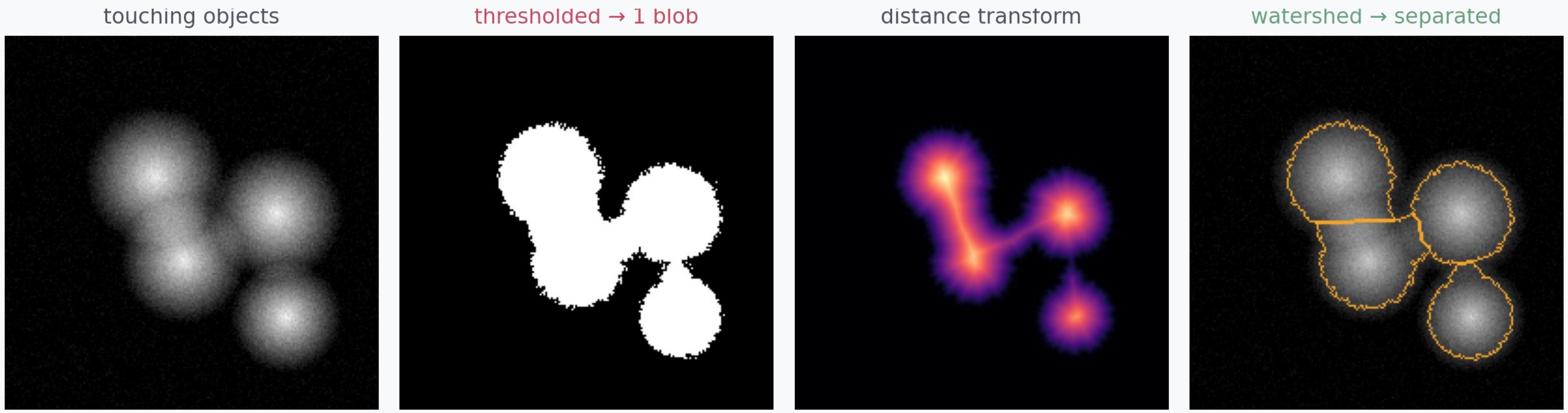
- Split: recursively divide the image into quadrants while a block is not yet homogeneous.
- Merge: glue neighbouring blocks back together once they turn out to share similar intensity.
- **No seed needed — the whole image is covered automatically.**



Quadtree partition (left) merged into a handful of homogeneous regions (right)

Watershed: flood the intensity landscape

- Treat the image as a 3-D landscape: height = intensity (or its gradient).
- “Flood” upward from each local minimum. Build a dam wherever two floods are about to meet.
- **Those dams are the segment boundaries — perfect for separating touching objects.**



KEY IDEA Flooding the raw image over-segments badly (every speck of noise is a basin) — in practice you flood a smoothed distance map seeded by a few markers.

Recap: region-based methods

- Region growing: simple, intuitive, but needs a seed and a tolerance.
- Split & merge: automatic and global, slightly blocky results.
- Watershed: excellent at separating touching objects — nuclei, cells, coins — once paired with good markers.
- All three respect spatial connectivity, fixing thresholding's biggest blind spot.

KEY IDEA Region-based methods ask “who are my neighbours?” — a question intensity-only thresholding never asks.

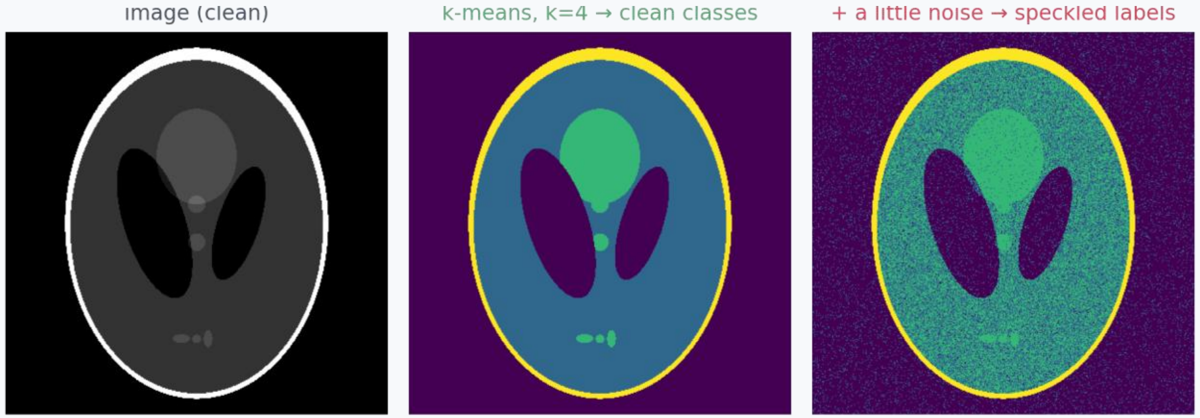
PART 4 OF 8

Clustering

What if a pixel's class depends on more than one number?

k-means segmentation

- Choose k (the number of tissue / object classes you expect).
- Iterate: assign each pixel to its nearest class mean → recompute each class mean → repeat.
- Minimises within-cluster spread — pixels in a class end up close to that class's mean intensity.
- **Like thresholding, but generalised to any number of classes, and the cut points are learned, not guessed.**



KEY IDEA k-means judges every pixel alone, exactly like thresholding — it has no idea that pixels next to each other usually share a label.

PART 5 OF 8

Statistical classification

Treat each tissue class as a probability distribution, not just a mean.

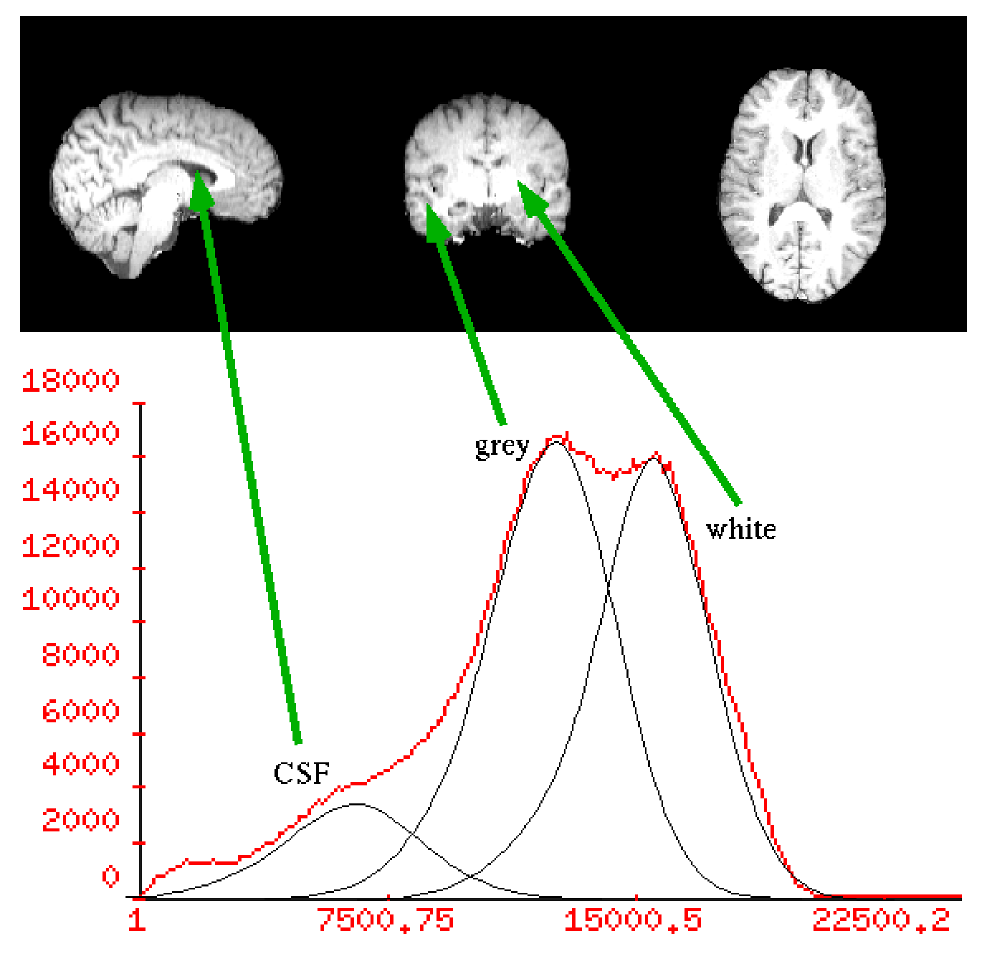
Reframing: segmentation as probability

- For each class (GM, WM, CSF, …), model the intensities pixels of that class tend to take — usually a bell curve (Gaussian).
- Given a pixel's intensity, ask: “under which class's bell curve is this value most probable?”
- Assign the pixel to that most-probable class.

KEY IDEA This is exactly k-means' idea of “nearest mean” — upgraded to also account for how spread out (uncertain) each class is.

The overlap problem

- Real tissue classes overlap heavily in intensity — grey and white matter especially.
- Wherever two bell curves overlap, a pixel could plausibly belong to either class.
- **Even tiny noise can flip the "most probable" class back and forth.**



Grey/white matter overlap so much that intensity alone keeps mislabelling pixels

ML vs. MAP: two ways to decide

Maximum Likelihood (ML)

Pick the class whose distribution makes this pixel's intensity most probable — using intensity alone.

Maximum a Posteriori (MAP)

Same idea, but also weigh in everything you already expect — e.g. that neighbours usually share a label.

KEY IDEA MAP is just ML plus a prior belief — and "neighbours agree" is the prior that motivates Part 6.

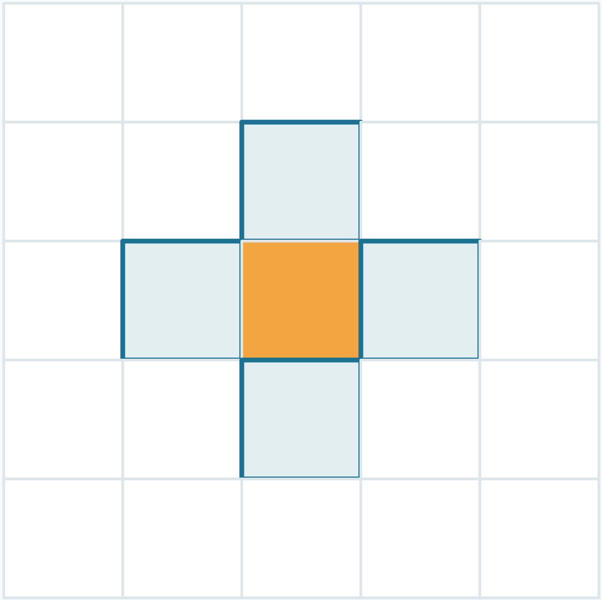
PART 6 OF 8

Markov Random Fields

Add the one prior that fixes almost everything: neighbours usually agree.

The key insight: spatial consistency

- Real anatomy is made of connected regions, not random speckle.
- If a pixel is grey matter, its neighbours are very likely grey matter too.
- **So: don't classify each pixel alone — classify it together with its neighbourhood.**



centre pixel + its 4 neighbours (a “clique”)

KEY IDEA This single assumption — smooth labels — is what separates a Markov Random Field from plain pixel-wise classification.

Turning the insight into a score

- Give every possible labelling of the image a score combining two penalties:
  - How well each pixel's label matches its own intensity (the statistical fit from Part 5).
  - How many neighbouring pixels disagree with it (a “smoothness” penalty).
- **Lower combined score = more plausible labelling. We search for the labelling with the lowest score.**

KEY IDEA That's all an “MRF energy” is: fit-to-data + disagreement-with-neighbours, added together.

Solving it: iterated conditional modes (ICM)

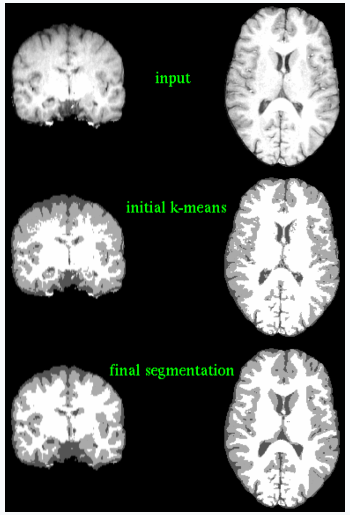
1
Start from any labelling (e.g. k-means)

2
Visit each pixel: try every label, keep whichever lowers the local score

3
Move to the next pixel

4
Repeat over the whole image until nothing changes

↻ this loop is exactly what alternates with re-estimating the class means — the same E-step / M-step idea you'll see again with the bias field in Part 7



Naive k-means gives noisy, speckled labels (middle). A few ICM sweeps clean it into anatomically coherent regions that respect tissue boundaries (bottom).

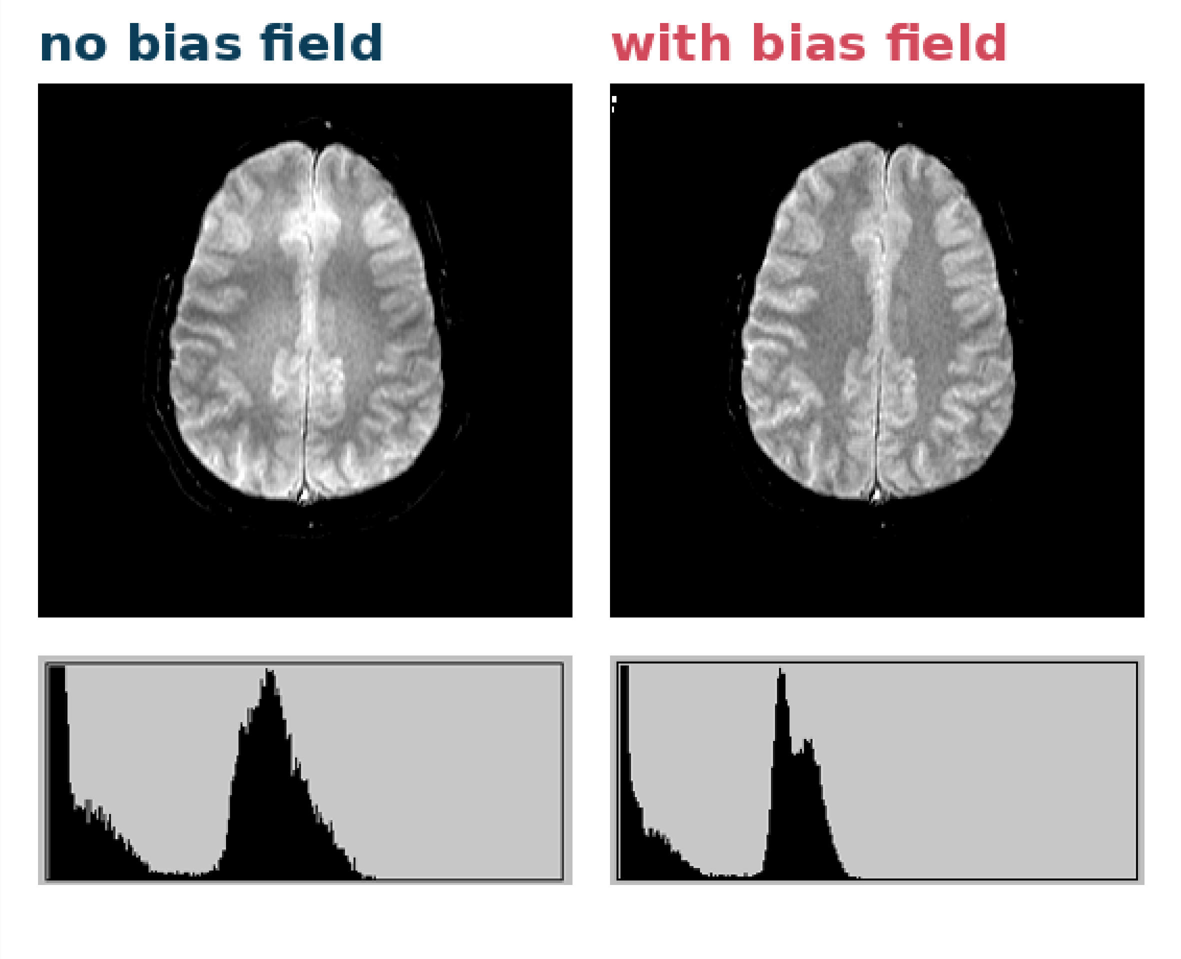
PART 7 OF 8

Correcting the scanner: bias field

Sometimes the obstacle isn't the anatomy — it's the acquisition.

The problem: a smooth, unwanted brightness gradient

- MRI coil sensitivity varies smoothly across the scan — the same tissue looks brighter on one side than the other.
- **This "bias field" smears out the histogram peaks that thresholding and clustering rely on.**



Identical anatomy, but the corrupted histogram's two tissue peaks have merged

Estimating the field and the labels together

E-step

Given the current estimate of the bias field, re-estimate the most likely tissue label at every pixel (MRF MAP step from Part 6).

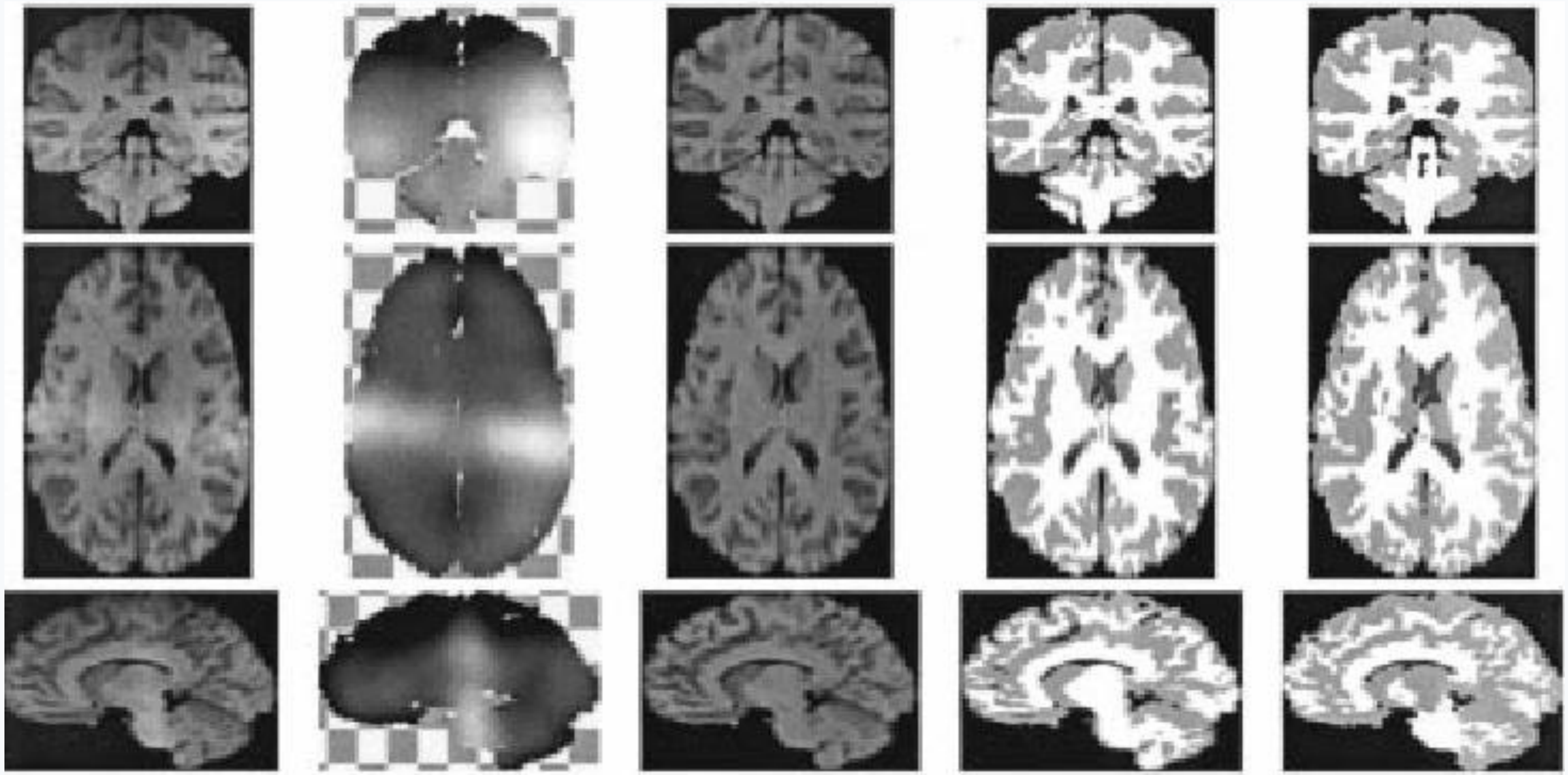
M-step

Given those labels, re-estimate a smooth bias field that best explains the remaining brightness pattern.

⟲ ***Alternate the two steps until the field and the labels both stop changing.***

KEY IDEA It's the same recipe as Part 6's ICM loop — just with a second unknown (the bias field) being refined alongside the labels.

The payoff: how close to a human expert?



Original — estimated bias field — corrected — automatic HMRF segmentation — manual expert segmentation

KEY IDEA Joint bias-field + MRF segmentation gets remarkably close to a skilled clinician's manual labelling — fully automatically.

PART 8 OF 8

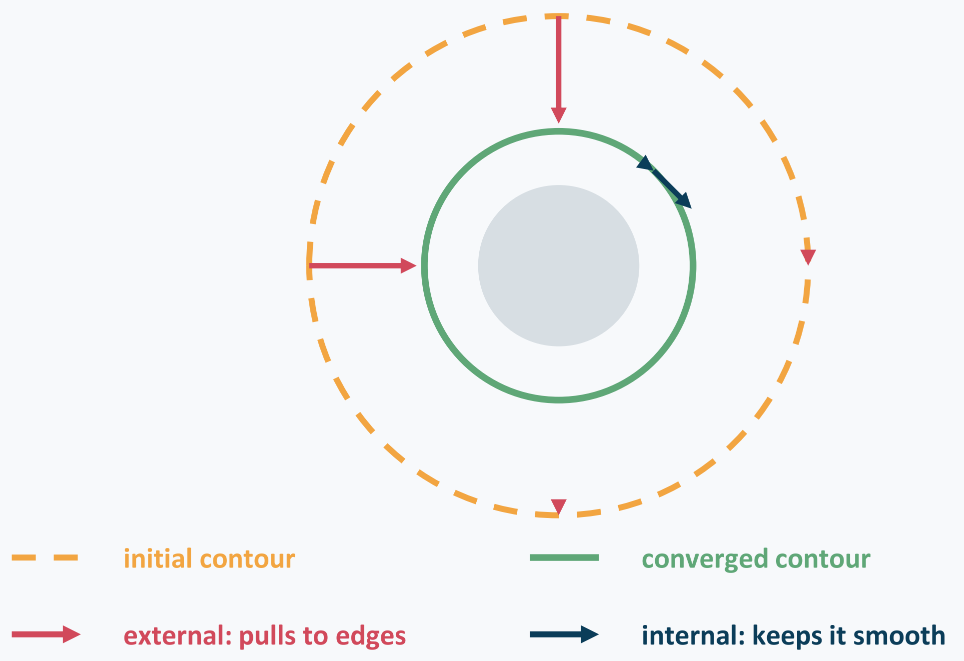
Deformable models: snakes

Stop labelling pixels — evolve a curve until it hugs the boundary.

Active contours: the rubber-band idea

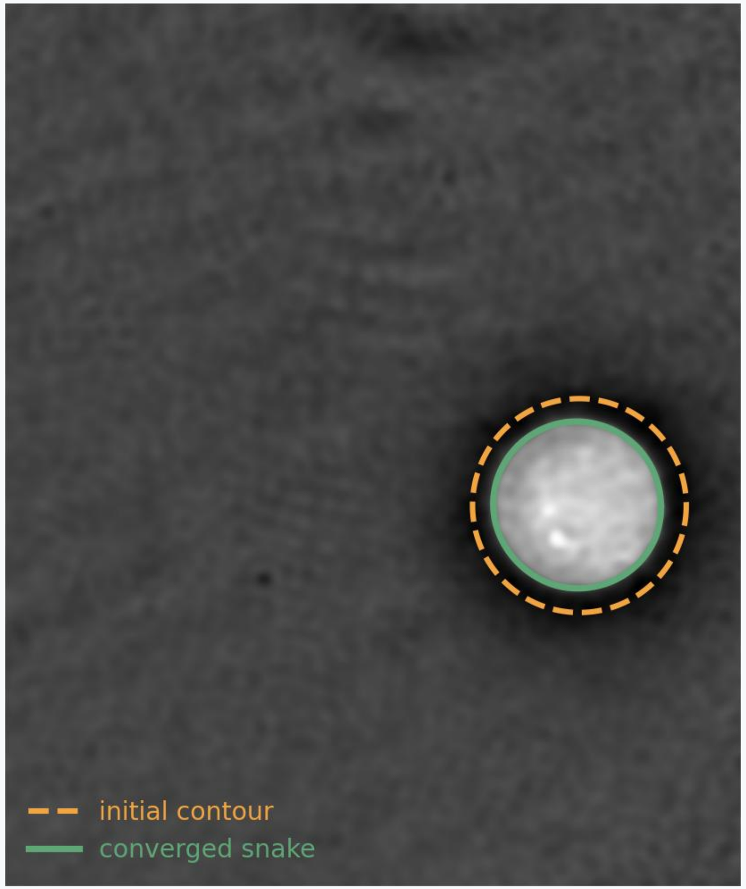
- Draw an initial closed curve roughly around (or inside) the object.
- Let it move like a stretchy elastic band, balancing two forces:
  - internal force — keep the curve smooth, resist kinking
  - external force — get pulled toward strong image edges
- **Iterate small movements until the curve settles — it has "converged" onto the boundary.**

Two forces acting on the curve



KEY IDEA No equations needed: a snake is just a smooth loop that an edge map quietly reels in.

Snake demo: from a rough circle to the cell boundary



Dashed initial contour shrinks onto the bright cell's true edge

Going further: level sets

- Snakes track explicit points on a curve — awkward when the shape needs to split or merge.
- Level sets instead represent the contour implicitly, as the zero-height slice of an evolving surface.
- Splitting, merging, and topology changes then happen automatically, for free.

KEY IDEA Same rubber-band philosophy as a snake — just a cleverer representation underneath.

Eight families, side by side

Family	Captures	Watch out for
Thresholding	Intensity alone	Uneven illumination
Edge-based	Sharp boundaries	Gaps, broken contours
Region-based	Connected, similar pixels	Seed / tolerance choice
Clustering (k-means)	Multiple intensity classes	No spatial context
Statistical / Bayesian	Probability per class	Overlapping class distributions
Markov Random Field	Statistics + neighbour agreement	Needs iterative solving
Bias-field correction	Smooth acquisition artefacts	Adds another unknown to estimate
Deformable models	Smooth object boundaries	Needs a reasonable initial curve

A practical classical pipeline

1
Pre-process
(denoise, bias-correct)

→

2
Segment
(pick the family that fits)

→

3
Clean up
(morphological open/close)

→

4
Validate
(compare to ground truth)

KEY IDEA Notice machine learning appears nowhere in this pipeline — and yet it still powers a great deal of real clinical and lab software.

FIVE THINGS TO TAKE HOME

1. Segmentation = classification: every method here is answering “which pixels belong together?”

2. Adding context beats adding cleverness — going from intensity → gradients → regions → neighbours steadily fixes each previous method's blind spot.

3. Otsu, Sobel/Canny, watershed and k-means remain genuinely useful pre-processing and QC tools, even inside modern ML pipelines.

4. A Markov Random Field is just “fit the data + agree with your neighbours” — no more mysterious than that.

5. Always ask what the acquisition did to your image (noise, bias field, partial volume) before blaming the algorithm.

Further reading

- Otsu, N. (1979). A threshold selection method from gray-level histograms. IEEE Trans. SMC.
- Canny, J. (1986). A computational approach to edge detection. IEEE Trans. PAMI.
- Kass, M., Witkin, A., Terzopoulos, D. (1988). Snakes: Active contour models. IJCV.
- Besag, J. (1986). On the statistical analysis of dirty pictures. JRSS-B. (ICM algorithm)
- Zhang, Y., Brady, M., Smith, S. (2001). Segmentation of brain MR images through a hidden Markov random field model and the EM algorithm. IEEE Trans. Med. Imaging.
- Sezgin, M., Sankur, B. (2004). Survey over image thresholding techniques. J. Electronic Imaging.

Thank you

Questions?